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ALIEN_X at the BUBT Bengali NLP Safety Challenge 2026: Label-wise Ensembles for Fine-Grained Toxicity Understanding

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Abstract

Bengali toxicity detection must account for indirect abuse, varied scripts, and attributes with few labeled examples. We present ALIEN_X's V38 submission to the BUBT Bengali NLP Safety Challenge 2026, which predicts overall toxicity and six related attributes. V38 combines BanglaBERT and lexical classifiers with earlier model outputs, a transliteration-adapted XLM-R encoder, and frozen Bengali sentence embeddings. Each label has its own fusion and decision rule. Using the historical 3,297/500/1,000 train/development/test split, we reconstruct six output columns from saved probabilities and match all corresponding test predictions. Development results show a clear trade-off for sarcasm: high recall comes with many false positives. The fitted implicit-toxicity stacker is missing, so we cannot verify a complete development score. ALIEN_X achieved a public leaderboard score of 0.608, which project records associate with V38. The team's private score of 0.591 is a separate result whose producing submission is not identified on the public page. Because the development data were reused during training and selection, these results describe the submitted system and its limitations rather than provide an independent estimate of generalization.

1 Introduction

Toxic language is not always explicit. A comment may threaten someone without using profanity, attack a group indirectly, or rely on sarcasm to convey abuse. In Bengali, these expressions also appear in Bengali script, romanized Bengali (Banglish), and mixed-script text. A useful classifier therefore needs both lexical cues and contextual information, while allowing different toxicity attributes to behave differently.

The BUBT Bengali NLP Safety Challenge 2026 captures this distinction through seven binary labels. Our V38 submission brings together predictions from several stages of the competition. It keeps four attributes from an earlier ensemble and changes the ranking rules for toxicity, implicit toxicity, and threats. The final decisions depend on this combination of models and post-processing, rather than on any one encoder.

We describe how V38 was assembled and examine what can be reproduced from the surviving notebooks and prediction files. We recover six output columns, report their development precision, recall, and F1, and discuss the errors behind those numbers. The paper is a retrospective system description; its comparisons should be read in light of repeated development-set use.

Split	Texts	Duplicates*	Bengali script†
Training	3,297	717	2,650
Development	500	102	408
Hidden test	1,000	198	792

Table 1: Historical V38-aligned data. *Repeated text rows beyond the first occurrence, as counted in the artifact audit. †Texts containing Bengali-script characters, including mixed-script texts.

2 Task Description

For an input text x, the system outputs $\hat{\mathbf{y}} \in \{0,1\}^7$. The labels cover overall toxicity, implicit toxicity, sarcasm, profanity, threats, identity attack, and sexual content. Several labels may be positive for one text. We evaluate the supplied annotations without adding a hierarchy that forces secondary predictions to agree with the toxicity prediction.

The official metric, confirmed against the competition evaluation page and implemented in the project, is [Hossain et al., 2026]

$$S = \tfrac{1}{2} F_{1,\text{toxic}}^{\text{macro}} + \tfrac{1}{2} \left(\tfrac{1}{6} \sum_{j=1}^{6} F_{1,j}^{+} \right), \qquad (1)$$

where toxicity macro-F1 averages the non-toxic and toxic class F1 values, whereas $F_{1,j}^{+}$ evaluates the positive class of secondary label j. A simple mean of seven positive-class F1 values is a different measurement. Per-label precision, recall, and F1 below refer to class 1 unless explicitly stated otherwise.

3 Dataset and Provenance

Table 1 summarizes the data associated with the saved V38 probabilities and submission. IDs are unique within each split, but many comments are repeated. After applying Unicode NFC normalization, collapsing whitespace, and trimming surrounding spaces, we find no exact text overlap between splits. This check does not rule out semantically similar comments. Within the labeled splits, 385 training groups and 63 development groups contain identical normalized text with conflicting labels. We retain the original labels: disagreement alone does not establish which annotation is correct.

Label frequencies are uneven (Table 2). Toxicity is pos-

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Label	Training positives	Dev. positives
Toxicity	2,357	353
Implicit toxicity	110	15
Sarcasm	566	88
Profanity	367	55
Threatening	93	15
Identity attack	1,379	211
Sexual content	159	24

Table 2: Positive-label counts. Labels can overlap.

itive in 71.49% of training rows, whereas threats account for 2.82%. Development contains only 15 positive examples each for threats and implicit toxicity, making estimates sensitive to a few decisions.

The archive also contains a different release with 3,292 training rows. Only 55 development IDs and 198 test IDs overlap with the historical probability files. We exclude that release and align the historical labels and probabilities by ID throughout the analysis. Matching these files by row position would give invalid results.

V35 used five grouped folds over all 3,797 labeled rows. The fold assignment contains 2,978 normalized-text groups, with no group split across folds. Grouping prevents identical normalized comments from appearing on both sides of a fold. However, these same data also contributed to component development and later selection, so the final development results are not an untouched holdout evaluation.

4 Related Work

BanglaBERT provides pretrained representations and language-understanding benchmarks for Bangla [Bhattacharjee et al., 2022]. It supplies the principal contextual model family in our earlier ensemble and several specialists. XLM-R learns multilingual representations at scale [Conneau et al., 2020]; V38 incorporates earlier XLM-R outputs and a transliteration-adapted checkpoint in its toxicity branch. BGE-M3 supports multilingual text embeddings [Chen et al., 2024]; a Bengali-adapted checkpoint supplies features for our threat classifier [Hasan, 2025].

These studies provide the representation families used in V38, but their published benchmark results do not establish performance on this competition. We assess the assembled system through its saved predictions. Our evaluation also follows the concern raised by Cawley and Talbot [2010]: repeatedly selecting models on a finite evaluation set can overfit the selection criterion. We therefore report development diagnostics separately from leaderboard scores.

5 System Overview and Methodology

Figure 1 summarizes the pipeline. Let $p_j^{(2)}$ denote the saved V2 probability and $p_j^{(14)}$ the V14 signal reconstructed from it and older snapshots. Before applying $\ell(p) = \log(p/(1-p))$, probabilities are clipped to $[10^{-5}, 1-10^{-5}]$. The logistic function is σ.

5.1 Lexical and contextual baseline

The saved V2 configuration selects csebuetnlp/banglabert, length 128, seed 42, learning rate 2×10^{-5}, and five epochs. Its probability ensemble weights BanglaBERT by 0.7 and a TF-IDF logistic-regression component by 0.3. The archived configuration records separate logistic-regression regularization strengths: $C = 4.0$ for toxicity, 0.5 for implicit toxicity and threats, and 1.0 for the other labels. This branch contributes the frozen V2 arrays; it is not retrained during V38 assembly.

Older `oldb` and `oldx` arrays supply BanglaBERT and XLM-R snapshot probabilities. Array indices below are zero-based. V14 toxicity combines $0.875p^{(2)}$ with 0.125 times the mean of older XLM-R snapshots 1 and 2. Its implicit signal is a 0.9/0.1 logit blend of V2 and the mean of older BanglaBERT snapshots. These dependencies distinguish V38 from the simpler saved V2 configuration.

5.2 Transliteration-adapted toxicity branch

V35 fine-tunes aplycaebous/tb-XLM-R-fpt using five folds and three epochs. The repository identifies the XLM-R architecture but provides no model card; the historical revision is not pinned [aplycaebous, n.d.]. It averages snapshots from epochs 2 and 3 and then fold predictions for the test set. Training uses script/label-weighted sampling and weighted multilabel binary cross-entropy. For logit z_{ij} and positive weight w_j, the loss has the form

$$\mathcal{L} = -\frac{1}{7N}\sum_{i,j}\big[w_j y_{ij}\log\sigma(z_{ij}) + (1-y_{ij})\log(1-\sigma(z_{ij}))\big]. \tag{2}$$

The final toxicity ranking signal is

$$q_{\text{tox}} = \sigma\Big(0.70\ell(p_{\text{tox}}^{(14)}) + 0.30\ell(p_{\text{tox}}^{(35)})\Big). \tag{3}$$

V38 labels the 673 highest-scoring test texts toxic, preserving the V14 positive count. Development reconstruction retains 340 positives. This is a batch-level rank decision, not a calibrated 0.5 cutoff.

5.3 Implicit-toxicity meta reranking

The implicit branch inherits V36's logistic meta model. Features include saved per-label probabilities, their logits, toxicity-context probabilities, and nine text features: log character/word counts, Bengali/Latin character proportions, exclamation/question-mark counts, and three script-category indicators.

The meta-model draws on MuRIL (V18), staged BanglaBERT and lexical outputs (V32), multilingual-E5 and Twitter-XLM-R (V33), V35 encoders, conditional attributes (V23), rare-label specialists (V15), and dedicated toxicity models. Appendix A lists the checkpoint families. These earlier models are dependencies of the implicit branch, even though they do not all appear as separate branches in Figure 1.

The reproduction notebook standardizes the input features and selects $C \in \{0.01, 0.03, 0.1, 0.3\}$ by five-fold out-of-fold average precision on the training rows. It then fits the meta-classifier on all 3,297 training rows. Some of its input probabilities, however, come from base models cross-validated over all 3,797 labeled rows. Fitting the stacker only on training rows therefore does not remove its dependence on development labels.

The meta-model probability is blended with $p^{(14)}$ in logit space, using $w \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Development F1 selects the blend, with fold-level wins and losses and the number of changed test decisions used to break ties. The

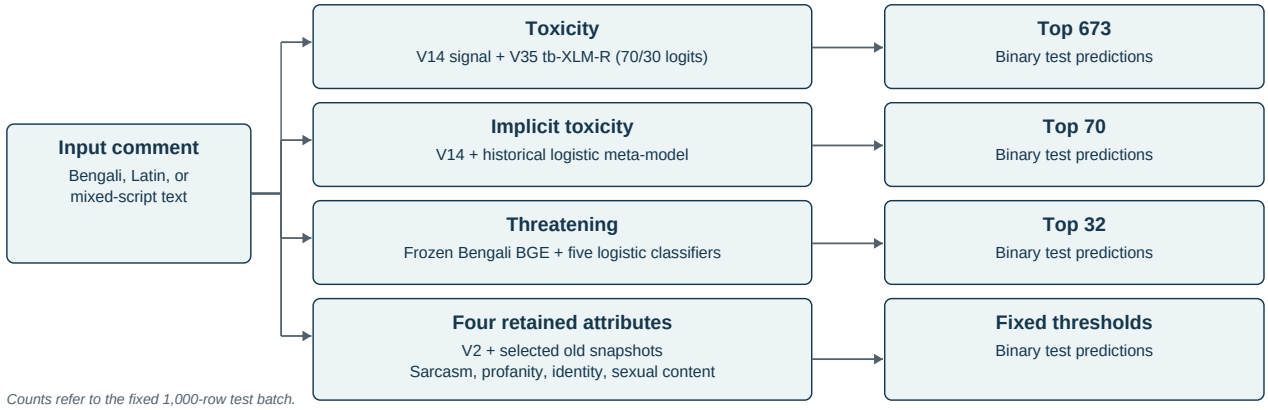


Figure 1: V38 inference from saved component probabilities. Four attributes retain V14 decisions. Three other labels use separate reranking branches. Counts apply to the fixed 1,000-row historical test batch; they are not hidden-label frequencies.

V35 setting	Recorded value
Maximum sequence length	128
Folds / epochs	5 / 3
Prediction snapshots	Epochs 2 and 3
Batch size / accumulation	4 / 4
Optimizer	AdamW
Backbone / head LR	1.2×10^{-5} / 2.5×10^{-5}
Weight decay / warmup	0.01 / 0.10
Fold seeds	2026, 17, 3407, 1337, 42

Table 3: V35 neural-component settings, not a universal configuration for every ensemble member.

procedure preserves 70 test positives and 28 development positives. Archived notes record $C = 0.01$ and $w = 0.40$, but the fitted scaler, coefficients, and final continuous scores are missing. We describe the recorded procedure without refitting the stacker or treating its development predictions as recovered.

5.4 Threat specialist

For threats, V38 uses frozen embeddings from nahidstaq/bge-m3-bangla with five logistic-regression classifiers. The recorded settings are `liblinear`, $C = 0.35$, and positive-example weighting capped at 3. The fold predictions are combined, and the highest-ranked examples are selected: 32 in the test set and 16 in development. Section 7 reports the resulting diagnostic, although the available experiments do not isolate the contribution of the embedding model itself.

5.5 Retained attributes and assembly

Sarcasm, profanity, identity attack, and sexual content retain V14's decision rules. Sarcasm uses $p^{(2)} \geq 0.175$, and sexual content uses $p^{(2)} \geq 0.245$. Profanity uses $0.95p^{(2)} + 0.05p^{(\mathrm{oldb},0)} \geq 0.245$, while identity attack uses $0.70p^{(2)} + 0.30p^{(\mathrm{oldb},2)} \geq 0.3814$. To assemble V38, the V36 target system's profanity column is replaced with V14's predictions. This changes 48 test decisions and leaves every other column unchanged. V37 is a different variant: it also differs from V38 on 12 threat decisions.

6 Experimental Setup

The training notebooks were written for Google Colab with CUDA support. They use Python, PyTorch, Hugging Face Transformers, Sentence Transformers, NumPy, pandas, and scikit-learn. The available records do not preserve a complete software environment or hardware log, so exact package versions, GPU model, peak memory, and total training time remain unknown. Table 3 gives the V35 settings; the V2 and meta-model settings are described separately because they come from different runs.

For this analysis, we align the saved V2 probabilities with the historical development IDs and rebuild V14 from the older snapshot arrays. We then apply the V36 toxicity and threat rankings and restore V14's profanity predictions. The six recoverable columns match all 6,000 corresponding decisions in the saved V38 test file. We use the same rules to compute development confusion counts without retraining the neural models. The implicit test column matches V36, but its development predictions are missing. This establishes agreement with the saved test outputs for six labels, while leaving full training reproduction unresolved.

7 Results

7.1 Development performance

Table 4 reports the six reconstructed outputs. The implicit-toxicity row is left unreported because its predictions could not be verified. Toxicity reaches positive-class F1 of 0.8745, compared with 0.7166 for the non-toxic class. The competition metric uses their macro-average, 0.795536. Threat F1 is 0.3871 when the 16 highest-ranked development examples are predicted positive. Both results are recomputed from the saved probabilities.

We can also reconstruct the complete V14 reference, which scores $S = 0.634443$. Its toxicity macro-F1 is 0.790835, and its implicit-toxicity positive F1 is 0.325581. Relative to V14, the V38 toxicity diagnostic rises by 0.004700, while threat F1 increases from 0.322581 to 0.387097. Each change reflects one additional true positive at the same number of positive predictions. The four retained attributes are unchanged. These small differences describe the selected systems on reused development data; they do not establish a general advantage for a model family. Without verified implicit predictions, we cannot report a complete V38 development score.

7.2 Leaderboard results and artifact comparisons

The official leaderboard lists ALIEN_X at 0.608 on the public split and 0.591 on the private split (Table 5), with 36 entries [Hossain et al., 2026]. Project records associate the

Label	Precision	Recall	F1 (positive)	Support	TP	FP	FN
Toxicity	0.8912	0.8584	0.8745	353	303	37	50
Implicit toxicity	—	—	—	15	—	—	—
Sarcasm	0.2195	0.9205	0.3545	88	81	288	7
Profanity	0.4889	0.8000	0.6069	55	44	46	11
Threatening	0.3750	0.4000	0.3871	15	6	10	9
Identity attack	0.5649	0.8246	0.6705	211	174	134	37
Sexual content	0.5556	0.6250	0.5882	24	15	12	9

Table 4: Cached-source reconstruction on 500 historical development rows. Support counts gold positives. Dashes denote unavailable implicit predictions, not zero performance. These are diagnostics after repeated selection, not hidden-test per-label results.

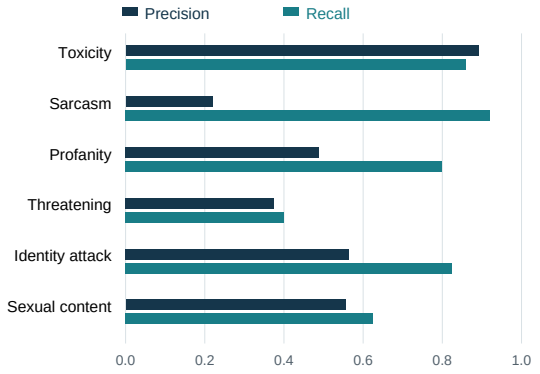


Figure 2: Precision and recall for six reconstructed development outputs. The unavailable implicit output is omitted.

ALIEN_X leaderboard	Score	Rank
Public (about 30% of test)	0.608	10
Private (about 70% of test)	0.591	7

Table 5: Team-level scores and displayed ranks on the completed Kaggle leaderboard, accessed October 12, 2026. These are not submission-specific score pairs.

public result with V38. The leaderboard itself does not identify the filename, hash, or private score of that submission, so we keep the team's private result separate from the V38 analysis.

Comparing the saved submission files shows exactly which decisions changed. V38 differs from V36 T595 on 48 profanity predictions and agrees on the other six columns. Relative to V14, it changes 36 toxicity, 40 implicit-toxicity, and 54 threat predictions. V39, in turn, differs from V38 only on 36 toxicity predictions. These comparisons document the submitted variants. Since test labels are hidden and the team leaderboard does not identify each variant, they cannot tell us which replacements improved test performance.

8 Error Analysis

Sarcasm has the clearest precision–recall imbalance. The classifier detects 81 of 88 positive examples, but also produces 288 false positives. It labels 73.8% of the development set as sarcastic, compared with a labeled prevalence of 17.6%. The low cutoff of 0.175 favors recall. Testing alternative thresholds on fresh data would help establish whether this trade-off is appropriate.

Among the secondary attributes with recovered predictions, identity attack has the highest F1 (0.6705), with 174 true positives and 134 false positives. Profanity finds 44 positive examples but produces 46 false positives, showing why its recall of 0.8000 should not be read on its own. Sexual content has a more balanced precision–recall profile, although only 24 positive examples support that estimate.

Implicit-toxicity errors cannot be quantified from the recovered V38 artifacts. Threat detection finds 6 of 15 positives and misses 9. One additional correct positive changes recall by approximately 0.0667 for either label. Small apparent changes therefore require caution. Toxicity misses 50 positive examples and flags 37 non-toxic examples. Its lower non-toxic F1 explains the gap between positive-class F1 and the official macro-F1.

The counts reveal different error balances across labels (Figure 2), but they do not explain the linguistic causes of individual mistakes. Determining whether code mixing, irony, or spelling variation accounts for those errors would require a separate review and annotation of the affected comments.

9 System Submission and Reproducibility

We analyze `submission_v38_NO_PN.csv`, the saved ALIEN_X submission containing 1,000 unique historical test IDs and seven binary columns. In task order, its positive prediction counts are 673, 70, 715, 180, 32, 618, and 63. These counts describe the system's decisions, not the prevalence of the hidden labels. Appendix A gives the file hash so the analyzed version can be identified unambiguously.

Code and notebook availability. The project repository is publicly accessible on GitHub.[1] Relevant entry points are the V2 baseline notebook, the V35 cross-script notebook, and the V36 assembly notebook. These document different parts of V38's development; later experiments in the repository are not necessarily components of V38. The paper package separately includes our aggregate audit report and reconstruction script, which requires the original V36 probability bundle and historical submissions. Raw comments, gold labels, and per-ID predictions are not included in the paper package. Repository access does not establish redistribution rights for every dependency. The fitted implicit stacker, full historical software environment, and exact neural checkpoint revisions remain unavailable, so the notebook links do not amount to a complete training reproduction. Appendix A lists the model families and submission hash.

The audit starts by verifying the 3,297/500/1,000 split and

[1] https://github.com/SamZx14/Bangali-Safety-NLP-Challange-Alien_X-2026, accessed October 12, 2026.

aligning probabilities by ID. It reconstructs V14, applies the V36 toxicity and threat rankings, and restores V14 profanity. The six resulting columns must match the saved test submission before development metrics are computed. Reproducing the training process would additionally require the missing checkpoint and stacker records; fitting replacements would constitute a new experiment.

10 Limitations

Repeated use of the development data is the main limitation. Some base predictions come from folds over the combined training and development sets. Development labels then influenced thresholds, blend weights, and candidate selection. An out-of-fold prediction at one stage does not make this entire process independent. The reported diagnostics therefore describe behavior on familiar data and should not be used to predict deployment accuracy.

The ranking rules also depend on the batch. Selecting 673 positives is well defined for this 1,000-example test set, but it does not provide a decision policy for an arbitrary stream of comments. A change in prevalence could make the fixed count unsuitable. Deployment would require thresholds or calibration evaluated in the intended setting. Duplicate comments and the small number of rare-label positives further limit what can be concluded here.

We have not conducted a demographic fairness study, an independently annotated error analysis, or a deployment evaluation. We also lack uncertainty estimates that account for duplicate groups and repeated selection. Missing software versions and unpinned checkpoints make exact retraining difficult. Further evaluation is needed before making claims about fairness or performance beyond the competition.

11 Ethical Considerations and Disclosures

A false positive can cause benign speech to be removed, while a false negative can leave harmful content unreviewed. The large number of sarcasm false positives is a particular concern for automatic moderation. Any practical use should include human review, an appeal process, and evaluation in the intended community. We report aggregate results and omit raw harmful comments and private user information.

The recovered V38 assembly uses competition labels and public pretrained models; it does not call a generative model to label test items. Available records do not demonstrate task-specific augmentation with external supervised examples, but incomplete historical artifacts prevent a categorical absence claim. Some source models use all 3,797 competition-labeled examples through cross-validation. No retrieval index appears in the recovered final inference.

Upstream model training also matters for disclosure. The current Bengali BGE model card describes augmented cross-script pairs and 12,000 synthetic pairs generated with Gemini 2.5 Flash and Claude Opus 4.5 [Hasan, 2025]. This is a statement about the released checkpoint's adaptation data; it is not evidence that ALIEN_X generated competition labels with those services. Because the historical revision was not pinned, the current card cannot fully establish which weights were used in V38. The card lists an MIT license, while the historical tb-XLM-R checkpoint's license documentation remains unresolved.

The authors report ChatGPT assistance with Python development/review, experimental design, evaluation, and pipeline optimization. AI assistance also supported manuscript drafting and figure preparation. Numerical claims remain grounded in supplied records and recomputed counts. No new neural training or leaderboard submission was performed for this paper.

12 Conclusion

V38 brings together lexical and contextual predictions through separate decision rules for each label. We recover six of its output columns and verify them against the saved test submission. Their development errors show why the aggregate score alone is insufficient: sarcasm recall is high but produces many false alarms, and rare-label results depend on very few examples. The team's public score of 0.608 is associated with V38 in project records; its private score of 0.591 remains a separate team-level result. The next steps are to evaluate on fresh data, reconsider the recall-heavy thresholds, and replace fixed prediction counts with policies suited to changing batches. Preserving fitted stackers, checkpoint revisions, and software environments would make those comparisons easier to reproduce.

Acknowledgments

We thank our supervisor, Md. Darun Nayeem, the BUBT Bengali NLP Safety Challenge organizers, and the researchers who released the pretrained models used in this work.

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A Checkpoint and Artifact Inventory

The identifiers below describe recovered notebook dependencies. They do not imply that every checkpoint was directly blended into every final label or that current weights reproduce historical ones.

- csebuetnlp/banglabert: baseline, older snapshots, attribute specialists, staged models, and toxicity context.
- xlm-roberta-base: earlier multilingual and toxicity branches.
- google/muril-base-cased: V18 multilabel probabilities.
- microsoft/mdeberta-v3-base: V19 toxicity context.
- csebuetnlp/banglabert_large: V25 toxicity context.
- intfloat/multilingual-e5-base: V33 embedding source.
- cardiffnlp/twitter-xlm-roberta-base: V33 contextual source.
- kazalbrur/bangla-embed-e5-small-banglish: V35 embedding source.
- nahidstaq/bge-m3-bangla: V35 threat embedding source.
- aplycaebous/tb-XLM-R-fpt: V35 toxicity reranking source.

The frozen V38 CSV has SHA-256 (concatenate the two lines):

```
17679bdcf8514ada3e6c2bb430f8d2f5
a460dd433a055e54580830b3b9858f10
```

The accompanying audit script and aggregate evidence report distinguish recomputed, reported, and unverified claims. Metric reproduction requires IDs, binary labels, and predictions, so raw toxic texts need not be redistributed.